Dongpil Kwak

Senior Native Mobile & On-Device AI R&D Engineer

Professional Overview

Senior Mobile R&D Engineer with over 15 years of experience specializing in native iOS and Android development, scalable system architectures, and on-device AI integration. Proven success in taking complex applications from concept to high-volume production, including founding engineering roles for market-leading logistics platforms (10,000+ daily deliveries) and delivering over 50 client-facing enterprise solutions within leading digital agencies.

Core Competencies & Technical Expertise

- **Native iOS & Android Architecture:** Extensive experience with **Kotlin, Coroutines, Hilt, Swift, async/await, Multi-threading**, and **MVVM** architecture to build highly performant, resilient native applications.
- **Scalable Architecture & CI/CD:** Architected and implemented the end-to-end mobile architecture and automated deployment pipelines (Fastlane, GitHub Actions, Firebase) for the Sherpa Driver platform from inception to national scale.
- **On-Device AI & Local LLMs:** Hands-on experience building and publishing local LLM AI Agents on Android (Google Play). Specialized in **Agentic Workflows**, **Retrieval-Augmented Generation (RAG)**, tool-calling, and on-device decision frameworks in mobile environments.
- **High-Impact Client Solutions:** Delivered major enterprise apps, including the official Sydney Royal Easter Show app (iOS & Android) and CEVA Logistics' vehicle tracking and delivery management platform for Mazda Australia.
- **AI-Assisted Engineering:** Power user of AI-paired programming tools (GitHub Copilot) and automated review pipelines to optimize code quality, velocity, and architecture standards.
- **R&D Background:** Strong foundation in software research and development from the South Korean technology industry, combining deep engineering rigor with forward-looking product innovation.

💡 Key Improvements Implemented:

1. **Technical Stack Explicitly Highlighted:** Seamlessly integrated **Kotlin, Coroutines, Hilt, Swift, async/await, Multi-threading, and MVVM** into a dedicated core competencies section.
2. **AI & Agentic Focus Refined:** Explicitly added **Agentic Workflows** and **RAG (Retrieval-Augmented Generation)** to highlight your specialization in cutting-edge on-device AI architectures.